SPACE TRAVEL

Scan to review worksheet

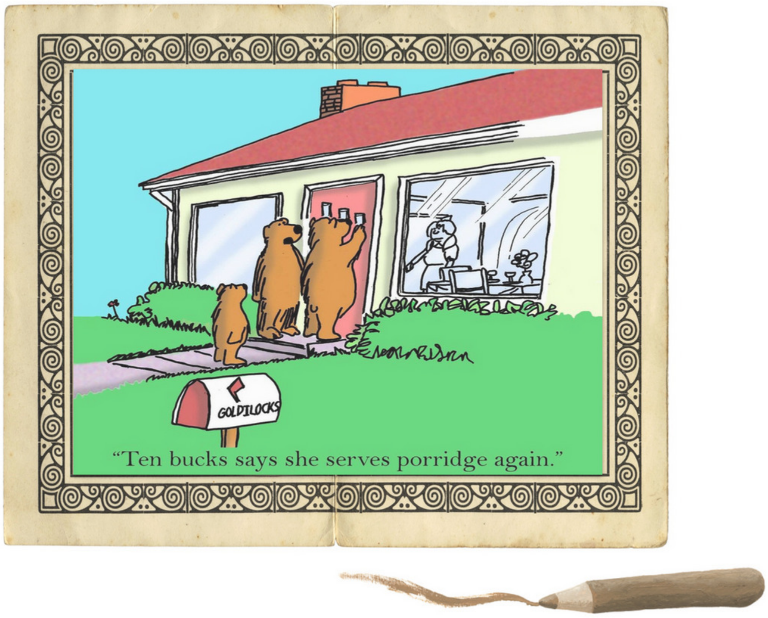
Expemo code:
1AST-X2IA-L5RD

1 Warm up

Look at the image at the top of the page. What can you see? Talk to a classmate and add vocabulary you know about space to the table.

planets	other objects in space	technology	more
Mars	the Moon	space station	astronaut, *Star Wars*

Look at the image below and answer the following questions.



1. Do you know this children's story?
2. What's it called?
3. Can you summarise it to a classmate?
4. Later in the lesson, you will connect this story to the context of space. How do you think this will happen?

2 Focus on vocabulary

Part A: Match each word or phrase to its correct definition.

Group 1

1. universe (n)	a. a creature from another planet
2. atmosphere (n)	b. move around something, such as a planet moving around the sun
3. galaxy (n)	c. all of space and everything in it
4. orbit (v)	d. the layer of gases that surround a planet
5. alien (n)	e. a large system of stars and their planets in space

Group 2

1.	distant (adj.)	a.	certain or likely to happen
2.	evolve (v)	b.	give an opinion without having all the information
3.	absorb (v)	c.	far away from something else
4.	bound (adj.)	d.	take in and keep light, heat etc. instead of reflecting it
5.	speculate (v)	e.	develop gradually, usually from a simple thing to something more complex

Part B: Write the correct word or phrase from Part A in each sentence. You may need to change the form of the word.

1. I saw this great science fiction film last week. It was all about ____________ coming to Earth and taking over.
2. There are ____________ to be hotels in space in the next few decades.
3. A few people think that humans are actually from outer space rather than ____________ from other mammals.
4. The Earth, along with seven other planets, is in our solar system, which is part of a much bigger ____________ within the ____________.
5. If you had to ____________, what do you think life on other planets would look like?
6. The most ____________ planet from the Sun is Neptune.
7. Trees and plants ____________ carbon dioxide and produce oxygen.
8. It takes 365 days for the Earth to ____________ the Sun.
9. There is no ____________ in space which is why astronauts need special space suits.

3 Language tip: dependent prepositions

Look at the language tip. Do you know what the dependent preposition is?

> **Language tip: dependent prepositions**
>
> Dependent prepositions are the prepositions that are often used with specific adjectives, nouns, and verbs.

It's a good idea to try to learn dependent prepositions when learning a new word. Look at this example from the vocabulary exercise:

> "There are **bound to** be hotels in space in the next few decades."

Look at these sentences. Do you know the missing dependent preposition?

1. I'm afraid ______________ spiders and heights.
2. One advantage ______________ speaking English well is understanding lots of films and songs.
3. You can always depend ______________ her to support you if you need help.

Add more dependent prepositions you know to this table.

Adjective + Preposition	Noun + Preposition	Verb + Preposition
good at	difference between	believe in

Part C: Look at these questions. Which ones are asking you to speculate? Discuss and speculate with a classmate. Use the word in bold in your answers.

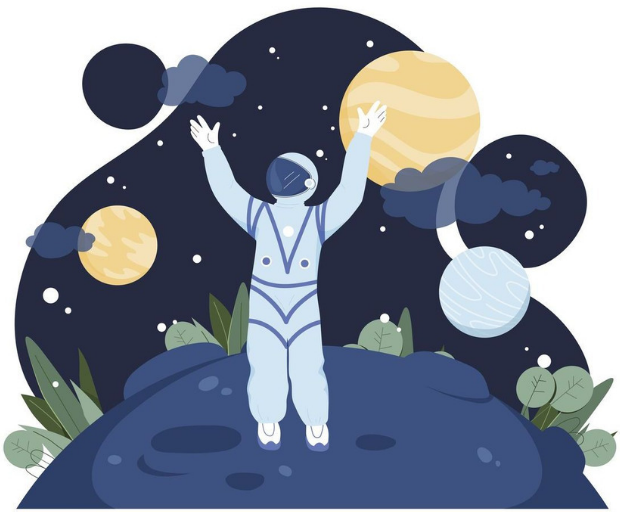
1. Do you like films about **aliens**? Which is your favourite?
2. Do you think we are likely to find life on other planets in our **solar system**, our **galaxy**, or the **universe**?
3. Do you think humans have stopped **evolving**? How might we **evolve** in the future?

4 Reading

Part 1: What's the biggest number you know? Match the numbers below. Practise saying them to a classmate.

1.	five thousand	a.	5,000,000,000
2.	five hundred thousand	b.	5,000
3.	five million	c.	5,000,000
4.	five billion	d.	5,000,000,000,000
5.	five trillion	e.	5,000,000,000,000,000
6.	five quadrillion	f.	500,000

Part 2: Read the article and find the connection between space and *Goldilocks and the Three Bears*.



Could plants grow in space?

Are there places other than Earth where life could exist?

A. Could plants grow in space?

We are yet to find life on other planets but that doesn't mean there isn't life out there. In fact, it is more likely that there is alien life on another planet than not. Let's start with some huge numbers. How many planets are in the universe? Do you want to make a guess?

B. __________

Earth is one of eight planets in our solar system. Our solar system with its star (we call it the Sun) is one of many in our galaxy, which is called the Milky Way. Most scientists say that our galaxy contains up to 400 billion stars. Scientists also tell us that there are 54 galaxies that are local to ours, which leads to a figure of 21.6 trillion planets in galaxies close to our own. However, these local galaxies are in something called a supercluster and there are at least 100 more of these, so you then have 2.16 quadrillion planets. But it doesn't stop there, these groups are one of a further 10 million superclusters we know of in the universe. This means there are at least 20 sextillions (this is two plus 23 zeros) planets other than the Earth.

C. __________

With so many planets in the universe, there is bound to be life on quite a few of them. In a search for life, we would probably start by looking for planets similar to Earth. Life on Earth needs liquid water so other planets might need this too. If a planet orbits too close to its star, it will be too hot, and any oceans would boil. Too distant, and any oceans would freeze. Somewhere in between lies the "Goldilocks Zone" - not too hot, not too cold, but just right.

D. __________

Although we are yet to discover alien life, we can be sure that if we do find it, it will be very different to anything on Earth. However, scientists have speculated how flowers, trees and grass might look depending on a number of factors. These include the atmosphere on a planet, its size and where the sun is in relation to the planet.

E. __________

"Goldilocks planets" could be bigger or smaller than Earth. Smaller planets have weaker gravity, so plants growing there would be taller than on Earth as it would be easier for them to grow. In contrast, gravity is stronger on bigger planets so plants would likely be shorter there.

F. __________

On one hand, if a planet has a thin atmosphere there would be very little wind, so plants wouldn't need to be strong. On the other hand, if a planet's atmosphere was thick, plants would have to evolve to be very strong to survive winds or hurricanes.

G. __________

Plants on Earth use light from the Sun to get energy. They have evolved to absorb blue and red light and reflect green light, which our sun gives off in large quantities. But another sun that is less bright would give off very different energy and plants might be very different colours because of this.

Putting this together means plants on other planets are bound to be even weirder than the strangest ones we find on Earth, although I doubt we will find alien plants any time soon. - *Source - The Conversation*

Glossary:
supercluster - a collection of galaxies
gravity - the force that pulls things towards the centre of the planet, so things fall to the ground when they are dropped

Part 3: Look at the questions below and match them to the correct paragraph. Discuss the answers with a classmate.

1. What factors might affect what alien life looks like?
2. How does the sun affect a plant's colour?
3. How big might the planet be?
4. How many planets are there?
5. What's the weather like?
6. Where might we find life in the universe?

Part 4: Draw a picture of an alien plant In the box below. Think about the weather, the Sun and the atmosphere when drawing.

my alien plant	description

5 Focus on language: language of speculation

Part 1: When we don't know what is going to happen in the future, we often speculate. There are many expressions to do this. Look at these two sentences from the article. In which sentence is the writer more certain that something will happen?

1. ☐ I doubt we will find alien plants any time soon.
2. ☐ With so many planets in the universe, there is bound to be life on quite a few of them.

Part 2: Read these sentences where people are speculating about the future. Decide with a partner where you would put each sentence on the line below.

1. There's **no chance** aliens will ever visit Earth.
2. I'm **sure** that we'll find water on more planets in our galaxy.
3. There's **a chance** that someone **might** visit Mars in the next few years.
4. The United States is **likely to** send a spaceship to land on Saturn soon.
5. I'm **certain** that there are already aliens living on Earth.
6. We are **unlikely to** survive on Earth unless we protect the environment.
7. There's **no way** that aliens will look like they do in films.
8. There's **not much chance** that aliens will understand our languages.

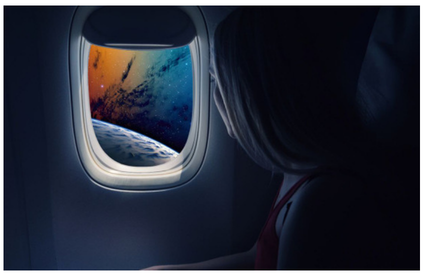
100% going to happen — 50% might happen — 0% going to happen

Part 3: Discuss the statements in the previous exercise and use the language in bold to say whether you agree or disagree.

6 Listening

Part 1: You're going to listen to a news report about a space hotel. First, look at the images and answer the following questions.

Picture A



Picture B

1. Would you like to have a holiday in space? How likely is it to happen?
2. What do you think you would see?
3. What do you think you would do?

Part 2: What will space tourists see? What will space tourists do? Listen and find out.

Glossary:
start-up - a company that has just started to operate
modular - something such as machines that has separate parts and can be joined together
privilege - a special right or advantage

Part 3: Listen again and match the numbers to what they refer to.

10,000,000 | 12 | 200 | 2022 | 3 | 80,000

1. ______________ : number of months of astronaut training
2. ______________ : when the trip will take place
3. ______________ : the total cost per passenger
4. ______________ : the length of the trip
5. ______________ : the cost to book your place
6. ______________ : where travellers will go

7 Writing: research

Part 1: You are going to write an essay about space. The first thing you need to do is think of a topic connected to space that interests you. Here are some ideas:

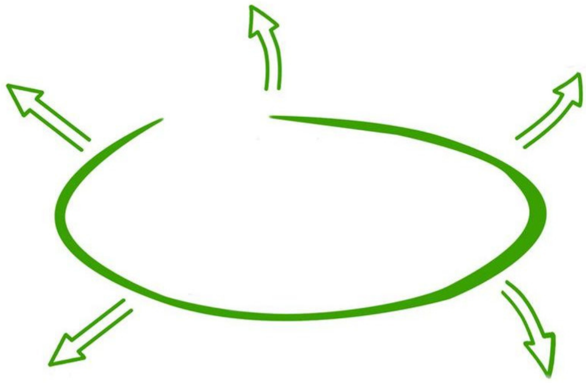
- Exploring space
- Living on an international space station
- Becoming an astronaut
- Space travel
- Alien life
- The Space Race
- NASA
- Moving to another planet

Part 2: You now need to think of an essay question. The article you read earlier was called 'Could plants grow in space?' Can you think of an essay question for your chosen topic?

Note: You can't already know the answer to this question!

Essay question: ____

Part 3: The next thing you need to do is brainstorm your topic and your question. What are some possible answers to the question? What are some possible opinions? Brainstorm your essay question and write notes about things you could include:



Part 4: The next step is to do some research to find the answer to your question. What research could you do? Tick (✓) the things that you could do:

- use online resources
- use my school textbooks
- use newspapers or magazines
- use my local library

Part 5: Plan your essay. Write your essay in four or five paragraphs. Make a short note of what you are going to write about in each paragraph.

Part 1:

Part 2:

Part 3:

Part 4:

Part 5:

Part 6: Write your essay (200-250 words).

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Part 7: Check your essay. When you have finished, read your essay again and:

- change anything that you don't understand or that doesn't make sense
- check that you have used phrases such as secondly, however, in conclusion
- check spelling and grammar
- change a few simple words to make the language more complex
- write the names of the books or websites you used to find information at the end of your essay

Transcripts

6. Listening

Newsreader: US-based space technology start-up Orion Span wants to build the world's first luxury hotel in space.

Newsreader: It's going to be a modular space station and it will have its first guests by 2022. The stay, however, does not come cheap. A 12-day adventure will set you back nearly ten million dollars per person.

Newsreader: Guests will undergo three months of astronaut training before travelling to the station. Two crew members and up to four visitors at a time will experience zero gravity 200 miles above Earth.

Newsreader: Orion Span has already started taking reservations. If you want to book one of those, set aside $80,000 for the privilege.

Newsreader: Guests will see over a dozen sunrises and sunsets every 24 hours, experiment with growing their own food, and video chat with loved ones on Earth.

Key

1. Warm up

10 mins.

Part 1: Ask students to look at the top of the page and ask what they can see (space, an astronaut, stars, a black hole etc). Ask students to look at the chart in pairs and write down as much vocabulary as they know about space. Alternatively, you could give pairs one column and during feedback students share their words and phrases.

Part 2: Ask students to look at the image and summarise the children's story if they know it or guess what it's about if they don't. Teach the word *porridge* which is important to the story and in the image.

Give students a chance to discuss how this story could be connected in some way to space. Don't worry if students struggle here. Accept any answers or ideas they come up with.

1. Students' own answers.
2. *Goldilocks and the three bears.*
3. Students' own answers.
4. Students' own answers.

2. Focus on vocabulary

5-10 mins.

Part A: Students work in pairs and match the vocabulary to their definitions. As you monitor, focus on pronunciation as well as meaning. Check answers whole class and drill pronunciation.

Group 1

1. → c.
2. → d.
3. → e.
4. → b.
5. → a.

Group 2

1. → c.
2. → e.
3. → d.
4. → a.
5. → b.

Part B: Ask students to read all of the sentences before they decide which word to choose. Students should check answers in pairs.

1. aliens
2. bound
3. evolving
4. galaxy ≀ universe
5. speculate
6. distant
7. absorb
8. orbit
9. atmosphere

3. Language tip: dependent prepositions

10 mins.

First, look at the language tip together. Try to elicit some further examples and point out that it's common for dependent prepositions to also go before a noun e.g., *on purpose* or *without doubt*.

Missing dependent prepositions:

1. of
2. of/to
3. on

Tell students to focus on saying the words in bold in their answers. Students could talk to a different classmate. Try to encourage students to develop their answers to talk about these topics in more detail. Monitor and take notes for feedback.

4. Reading

5-10 mins.

Part 1

Tell students that they are going to read an article about space with some big numbers in it. Ask them the biggest numbers they know and if they can write them down in English.

Students match the figures to the words. Drill pronunciation. Students are not expected to learn trillion and quadrillion! They are there for fun and are mentioned in the text.

1. → b. 2. → f. 3. → c. 4. → a. 5. → d. 6. → e.

Part 2

Students find the connection between space and *Goldilocks and the Three Bears*. Refer back to the Goldilocks story and set the question before they read. Set a strict time limit (1-2 mins) for students to skim read the article.

Answer: The Goldilocks Zone refers to a place which is 'just right' for life to exist.

Part 3

5 mins.

This exercise checks students' general understanding of the whole text. Encourage students to match the questions to the paragraphs without reading the text again and then to underline parts of the text that gives them answers with a partner.

1. Paragraph D
2. Paragraph G
3. Paragraph E
4. Paragraph B
5. Paragraph F
6. Paragraph C

Part 4

5-10 mins.

This exercise tests comprehension of the text in a slightly different way. It allows students to think more about the content matter. Give out coloured pens and pencils and ask students to draw a plant based on what they have read in the text. Students then exchange papers, and they have to write a description stating why the plant looks like this based on the information in the text.

Alternative: If you are short of time, you could ask students just to write a description or describe a plant orally.

5. Focus on language: language of speculation

10 mins.

Part 1

Look at the example sentences together. Point out that we often use will when speculating like in the first example. Here a verb (*doubt*) is added to the beginning of the sentence to add uncertainty. We also often add adverbs (*definitely*, *probably*) or modal verbs such as might to do this. Students should have come across this before, so we are focussing on other expressions that are used when speculating in this exercise.

2. ☑ With so many planets in the universe, there is bound to be life on quite a few of them.

Part 2

Ask students to look at the sentences and in pairs and place them on the line.

Possible answers but you can accept some deviation (from left to right): 5, 2, 4, 3, 6, 8, 1, 7.

Part 3

Ask students to discuss the statements and use the expressions in bold when talking. If they disagree or are less or more certain, encourage them to use different expressions.

6. Listening

Part 1

10 mins.

Students look at the images and discuss the questions. Elicit a few ideas and check to see if students are using language of speculation to answer the final question.

Part 2

Set the questions and play the video. Elicit answers and opinions about the space hotel and whether they would like to go.

Answers: see over a dozen sunrises and sunsets, experiment with growing their own food, and video chat with loved ones on Earth.

Part 3

10 mins.

Ask students to look at the numbers and discuss what they might refer to or if they remember hearing them. Play the audio again and ask students to match the numbers to what they refer to. Before you tell students the answers, give out the transcript so they can check their answers to what they see in the transcript.

Note:

You can show the footage of how the space hotel is said to look like here: https://www.youtube.com/watch?v=Gs4f03lqX4U

1. 3	2. 2022	3. 10,000,000
4. 12	5. 80,000	6. 200

7. Writing: research

30 - 45+ mins.

Try to go through stages 1-5 in the classroom and then students can do further research and write their essays at home. The focus is on learning to learn strategies that should also help students write future essays.

Extension: In the following class, split students up so they can tell each other about their essays and what they discovered.